

Battery Module Technical Specification and Validation Report

BM-51 | Revision C | Production release

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Prepared by	Fictional Energy Systems Engineering Group
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Document control

This report defines the released configuration and records the design rationale and validation evidence for the fictional BM-51 rechargeable battery module. Revision C is the current production release described by the specification clauses in this document. Values from Revision A and Revision B are retained where useful to explain design development and validation history. Unless a clause explicitly identifies another revision or a test sample, a requirement in this report applies to the Revision C production configuration.

The report is intended to support integration planning, engineering review, and controlled verification. It is not an installation guide, service manual, shipping declaration, or certification record. System integrators remain responsible for pack-level protection, wiring, thermal design, enclosure integration, and applicable regulatory review.

Document conventions

Requirements use shall for mandatory design or acceptance conditions. Should identifies a recommended practice. A value described as *nominal* is a design reference and is not a guaranteed limit. Minimum and maximum values are limits within the stated conditions. Test measurements describe particular samples and procedures; they do not replace the corresponding specification or acceptance criterion.

Temperatures are in degrees Celsius, current in amperes, voltage in volts DC, and energy in kilowatt-hours unless a table states otherwise. A dash indicates that the value is not applicable or was not recorded; it does not imply zero. Page numbers may change when the controlled PDF is composed. Section IDs are the stable references for requirements and evidence.

1 Introduction

1.1 Purpose

This document establishes the technical envelope and validation status of the BM-51 battery module. It brings together design characteristics, integration constraints, safety requirements, operating conditions, and selected verification evidence. The intended reader is an engineering team assessing whether the module is suitable for a fictional low-speed industrial mobile platform or stationary laboratory demonstrator.

The report distinguishes three kinds of information: released design requirements, recommended integration practices, and results from identified validation activities. A measured value is evidence about the tested sample and procedure. It is not automatically a guaranteed production limit. Similarly, an engineering note may explain a design choice without creating a requirement.

1.2 Scope and boundaries

The BM-51 is a removable, enclosed energy-storage module built around series-connected lithium iron phosphate cells. The module contains cell interconnects, sensing harnesses, a monitoring controller, contactors, fusing, and a liquid-cooled base plate. The external cooling loop, vehicle inverter, charger, supervisory controller, and system-level crash protection are outside the module boundary.

This report addresses the module as an assembled product in the Revision C production configuration. It does not define cell manufacturing controls, charger algorithms, vehicle-level functional safety, or pack parallelization rules. Integration conditions that affect module behavior are identified in Section 7. System-level designers must treat the module limits as inputs to their own hazard analysis.

1.3 Configuration and applicability

The controlled product is identified by the fictional family code BM-51 and a configuration suffix on the rating label. Revision C is the released production configuration. Its electrical and mechanical design is described in Sections 3 and 5; operating limits and safety conditions are in Sections 6 and 7.

The earlier configurations are included for design traceability:

Revision	Configuration status	Distinguishing context
A	Early design study	Preliminary electrical concept. Not released for integration or validation acceptance.
B	Prototype / validation configuration	Used for design verification and engineering trials. Some results remain relevant as evidence, but B-specific ratings do not define the C release.
C	Current production release	Configuration controlled by this report. Requirements identified as released specifications apply to C.

Hardware serial numbers and test records identify the revision actually tested. A result from a Revision B sample may support a design conclusion where the tested feature is unchanged, but it cannot silently override a Revision C requirement. Any such use requires a documented applicability assessment.

1.4 Terms and abbreviations

Term	Meaning in this report
BMS	Battery monitoring system, including sensing, state estimation, diagnostics, and protection commands
EFC	Equivalent full cycle; accumulated discharge throughput normalized to rated usable capacity

SOC	State of charge, expressed as a percentage of the controller's estimated usable range
SOH	State of health estimate based on capacity and resistance indicators
Cell temperature	Temperature measured at the defined cell-surface sensor locations; not coolant temperature
Module boundary	Electrical and mechanical interface at the module connectors and mounting features

2 System Overview

2.1 Functional description

The BM-51 stores electrical energy for a host system and supplies a nominal low-voltage DC bus. The BMS monitors cell-group voltage, current, and selected temperatures. When commanded by the host or when a protection condition occurs, the BMS can open the internal contactors. A service disconnect isolates the cell stack for maintenance procedures; it is not a substitute for verifying absence of voltage.

During charging, the host charger controls the current and voltage profile. The module reports readiness and constraint information over a CAN-based interface. It does not independently regulate the external charger. The host shall honor charge and discharge limits reported by the module and shall independently prevent operation outside the conditions in Section 7.

2.2 Electrical and thermal architecture

The cell stack uses sixteen series groups. Parallel cell count and internal interconnect geometry are controlled manufacturing details and are not a field-replaceable configuration. A fused positive path and two-pole contactor arrangement separate the cell stack from the external terminals. Voltage sensing is routed independently from the main current conductors to reduce measurement error under load.

Heat generated in the cells and conductors is transferred through electrically isolated thermal pads to an aluminum base plate. Coolant passages are internal to the module base. The module contains no pump, radiator, or coolant reservoir. The host thermal loop supplies conditioned coolant and monitors inlet and outlet conditions. Refer to Section 4 for thermal limits and Section 7.3 for external-loop requirements.

2.3 Interfaces

The module has a keyed DC power connector, a low-voltage signal connector, two coolant ports, and four mounting points. Connector pin assignments are defined by the separate interface-control drawing associated with the fictional BM-51 project. The interface drawing is not reproduced here; this report provides performance constraints rather than a complete wiring definition.

The CAN interface transmits estimated SOC, allowable current, temperature status, and diagnostic state. Communication loss does not authorize continued charging at a previously received limit. The host shall transition to its defined safe state after the configured communication timeout. The module's own protection remains active, but host-level shutdown behavior must be validated in the integrated system.

2.4 Configuration distinctions

Revision B prototypes used a different busbar and controller calibration. They were appropriate for thermal mapping and early vibration development but did not have the final production sensing harness. Revision C introduced a revised busbar, updated SOC estimator, and production-intent seals. Where test results depend on these changes, this report identifies the limitation in Section 8.

The accessory coolant manifold used in one laboratory fixture is not part of the module. Its connector body was rated IP65 as a component. That component rating does not define the ingress protection of the assembled module; see Section 5.5 for the module enclosure condition.

3 Electrical Characteristics

3.1 Electrical Ratings

The released electrical ratings in this chapter apply to the Revision C module at the external DC terminals. They assume the environmental and coolant conditions in Section 7, a functioning BMS, and correctly installed high-voltage interlocks. Ratings are not permission to exceed a limit briefly unless a duration is explicitly specified.

Characteristic	Revision C production specification	Notes
Series cell-group count	16	Internally connected series groups
DC terminal polarity	Keyed, fixed polarity	Confirm markings before connection
Nominal terminal voltage	51.2 V DC	Design reference; operating voltage varies with SOC and load
Maximum continuous charge current	50 A	Subject to lower BMS-reported limit and thermal conditions
Maximum continuous discharge current	100 A	Continuous rating, not a pulse rating
Short-duration discharge capability	120 A for no more than 10 s	Followed by at least 60 s below the continuous limit

The 120 A allowance is a transient capability and must not be interpreted as a revised continuous rating. The host shall enforce the current-duration envelope and shall not accumulate repeated pulses without allowing the module to return below its continuous current limit.

3.2 Rated Energy and Capacity

For the Revision C production configuration, the rated nominal energy is 5.12 kWh. This value is calculated from the production design's nominal voltage and rated capacity. Actual available energy depends on temperature, age, current profile, balancing state, and the BMS-defined SOC window. The nominal rating is not a guaranteed deliverable energy under every load profile.

The usable SOC window for normal service is 10% through 95%, inclusive of the stated endpoints for controller reporting. The BMS retains reserve outside this window to support cell protection and estimation stability. Cell-level voltage allowances extend beyond the normal module SOC window and must not be used by a host controller to enlarge usable capacity.

Revision A design notes list 4.80 kWh as a preliminary estimate based on a smaller cell format. That value was not carried into the production design. During one Revision C capacity characterization, the sample delivered 5.06 kWh under the laboratory profile in Section 8.4. The measurement is a test result, not a replacement for the rated nominal value; test temperature, current profile, and endpoint behavior affect the result.

3.3 Voltage limits and SOC interpretation

The BMS estimates SOC using coulomb counting corrected by voltage and temperature models. The displayed SOC is a software estimate, not a direct measurement of stored energy. Following a rapid change in load, estimator settling can be delayed; a known indication behavior is documented in Section 9.2.

The cell supplier's characterization envelope includes operation down to 5% and up to 100% cell SOC under controlled conditions. Those cell-level limits are not the BM-51 normal usable range. The module controller

reserves margin to account for aging, imbalance, measurement uncertainty, and protection response. Host software shall use the module-reported SOC and limits rather than reconstructing an expanded range from cell data.

3.4 Continuous Current Limits

The production continuous discharge rating is 100 A. It assumes the module inlet coolant and ambient conditions remain within the ranges in Section 7, connectors remain within their specified contact resistance, and the module is not in a derated state. The BMS may report a lower allowable current as cell voltage, temperature, or SOH approaches an operating boundary.

An engineering transient allowance permits 120 A for up to 10 seconds, provided the current then remains below 100 A for at least 60 seconds. The transient allowance does not supersede the continuous rating and does not apply if the BMS reports a lower limit. A Revision B prototype test used 120 A in a 15-second pulse while characterizing the earlier busbar. That test was conducted for characterization only; the longer duration is not a production capability.

3.5 Charge Limits

The maximum continuous charge current for Revision C is 50 A. The host charger shall reduce current when requested by the BMS and shall terminate or suspend charging when the module reports a fault or charge inhibit. Below 0 °C ambient, charging is prohibited by the module operating envelope even if the cells are electrically capable of accepting a small current. See Section 7.1.

The acceptance cycling campaign used 40 A to reduce test duration while maintaining a controlled thermal profile. This was a selected test current, not a 40 A production limit. Revision B firmware had an experimental 60 A charge setting; that setting was removed after temperature and connector-margin review and is not applicable to the released configuration.

3.6 Electrical monitoring and communications

Cell-group voltage, terminal current, and sensor temperatures are sampled by the BMS. Diagnostic limits use filtered measurements with defined debounce intervals. Host displays may update more slowly than internal protection decisions. An external instrument connected at the terminal may therefore record a transient that is not identical to the BMS's filtered value.

The CAN message set includes signal validity and freshness indicators. If a value is marked invalid or stale, the host shall not substitute an assumed permissive value. The detailed timeout and message counter behavior are specified in the interface-control drawing and integration test plan.

3.7 Electrical integration notes

The module shall be connected through a host circuit that provides suitable overcurrent protection and precharge behavior for the load. Cable and busbar sizing must account for temperature rise, installation bundling, and voltage drop. The module's internal fuse protects the module assembly; it is not necessarily sized to protect every external conductor under every fault arrangement.

4 Thermal Management

4.1 Heat paths and control strategy

Cell and interconnect losses are conducted through dielectric thermal pads into the base plate. The host cooling loop removes heat through the base-plate passages. The BMS uses multiple cell-surface sensors and base-plate sensing to estimate thermal state. Coolant temperature alone is not a substitute for cell temperature measurement.

The host should supply coolant flow before enabling sustained high-current operation. Thermal control should avoid rapid inlet-temperature changes that can create localized gradients. The module is not designed to reject full continuous-current losses by natural convection in a sealed installation.

4.2 Coolant interface conditions

Use a water-glycol mixture compatible with aluminum passages and the specified seals. Coolant chemistry, filtration, and service interval are controlled by the host system. The module ports are not self-sealing; coolant handling procedures must prevent trapped air and leakage during connection.

Interface condition	Engineering guidance
Recommended coolant inlet range	15–35 °C during sustained high load
Minimum flow for rated continuous discharge	4.0 L/min
Maximum loop pressure at module inlet	150 kPa gauge
Coolant leak check	Perform before electrical enable

These loop conditions support the rated envelope but do not change the ambient operating limits in Section 7.1. Host thermal design must account for pump failure and blocked flow. The module can derate or disconnect in response to detected temperature, but host controls should not rely on protective shutdown as a normal temperature-regulation method.

4.3 Thermal Control Limits

The maximum permitted cell temperature during operation is 60 °C. The BMS begins reducing allowable current before the maximum is reached, with derating dependent on temperature rate, sensor spread, and operating mode. A normal control target near 45 °C is used by the integration example to preserve margin; it is an operating target, not the maximum allowed cell temperature.

The hard-stop logic is validated against sensor uncertainty and thermal lag. A measured shutdown at 58 °C on one instrumented test sample was the result of that sample, sensor placement, and filtering. It is not the specified maximum cell temperature. The applicable production requirement remains the limit stated above.

4.4 Temperature Uniformity Requirement

At steady continuous load under the specified coolant conditions, the difference between the hottest and coolest monitored cell locations shall not exceed 8 °C. The value is a production design requirement for thermal uniformity, assessed after the defined stabilization period in the validation plan.

Revision B thermal mapping recorded a 10 °C spread at a 90 A discharge condition. The result prompted a base-plate flow-path change. Revision C validation later measured a 6.2 °C spread in the corresponding mapped condition. The latter is evidence from a test run; the acceptance threshold remains the limit defined in this subsection.

4.5 Thermal fault response

Loss of coolant flow, sensor disagreement, or a temperature rise inconsistent with the commanded current may cause a current limit or contactor opening. The host shall treat a thermal fault as a system-level event and shall not automatically re-enable the module until the fault cause is resolved and the BMS indicates readiness.

Inlet and outlet temperatures should be logged during commissioning. If the outlet-to-inlet differential increases unexpectedly, check flow restriction and trapped air before changing current limits. The diagnostic should include ambient temperature and load history so that transient measurements can be interpreted correctly.

5 Materials & Mechanical Design

5.1 Enclosure architecture

The enclosure consists of a formed aluminum tray, bolted upper cover, internal insulating barriers, and a machined cooling base. Dissimilar-metal interfaces use controlled coatings or isolating hardware to limit galvanic corrosion. Internal cell restraint maintains compression through the expected service temperature range; the restraint is not a field adjustment.

The cover provides access for controlled factory assembly and inspection. Opening the enclosure outside an authorized service process can compromise sealing, electrical spacing, and warranty traceability. No user-serviceable cell replacement is defined by this report.

5.2 Materials and finishes

The primary enclosure alloy is selected for stiffness, corrosion resistance, and thermal transfer. Busbars use a conductive copper alloy with plated contact surfaces. Polymer barriers are selected for electrical insulation and temperature capability. Final material substitutions require a change review because a nominally similar polymer or finish can alter flame behavior, creepage, corrosion, or sealing performance.

Material declarations for the fictional production bill of materials are maintained in the project configuration record. This report does not claim compliance with any real jurisdictional materials regulation. Integrators must perform their own materials and end-of-life assessment.

5.3 Mass Properties

For Revision C production, the module mass shall not exceed 42.0 kg, including the enclosure, internal hardware, and coolant passages in the dry state. It excludes coolant retained in the external host loop and external brackets. The installation structure should support the module and expected dynamic loads with appropriate margin.

The first Revision B enclosure assembly weighed 39.5 kg because it used a lighter prototype cover and omitted two production restraint features. A Revision C sample measured 41.6 kg during incoming validation. The sample measurement is consistent with the maximum limit but does not replace lot-level verification or the production mass requirement.

The center of gravity is located near the geometric center of the base plate when the module is fully assembled. Orientation marks identify the preferred installation direction. The host should avoid mounting configurations that place sustained bending load on the connector panel.

5.4 Mounting and mechanical interfaces

Four M8 mounting points attach the module to the host structure. Use fasteners with adequate engagement and prevent loosening under vibration. Do not use the coolant ports or electrical connectors as lifting or restraint points. The module should be supported across the mounting pattern to avoid twisting the cooling base.

Lifting fixtures used in factory handling engage dedicated temporary features. These features are removed or capped before installation and are not vehicle mounting points. Field lifting methods must account for the module mass and local workplace requirements.

5.5 Enclosure and Environmental Protection

The assembled Revision C module achieves IP67 when all specified seals, connector caps, and service closures are correctly installed. The rating applies to the enclosure configuration and test basis recorded in the validation plan. It does not establish resistance to every chemical, pressure-wash procedure, or prolonged immersion condition.

The external signal connector body is an IP65-rated component before mating. Its component rating is not the module rating and does not establish the assembled interface's protection unless the correct mating connector and seal are fitted. A laboratory accessory manifold used with the coolant rig carried a separate IP6K9K marking; it is not part of the BM-51 assembly and that marking is not applicable to the module.

An early unsealed lab fixture leaked during immersion exposure. That configuration lacked the production cover gasket and is not representative of the Revision C enclosure. Seal condition and connector closure must be checked after service or mechanical repair.

5.6 Handling and inspection

Inspect the enclosure for dents near the cooling passages, damaged connector keys, displaced seals, and evidence of coolant leakage before installation. A damaged enclosure shall be quarantined for engineering review. Do not energize a module with visible liquid ingress or an unexplained insulation fault.

6 Safety Requirements

6.1 General precautions

The module contains stored electrical energy even when external contactors are open. Qualified personnel shall follow the host system's lockout procedure, verify absence of hazardous voltage with an appropriate instrument, and use insulated tools where required. The service disconnect reduces exposure but does not eliminate all internal electrical hazards.

The module is not a complete battery system safety solution. The integrator is responsible for system enclosure, crash protection, emergency shutdown, fire response planning, conductor protection, and operator procedures. Risk assessment must consider foreseeable misuse, component failure, environmental exposure, and maintenance activity.

6.2 Electrical Isolation

For the assembled production module, insulation resistance from live circuitry to the conductive enclosure shall be at least 500 Ω/V when tested at 500 V DC after the specified stabilization interval. The requirement applies with the module in the test configuration defined by the validation procedure and with external interfaces isolated as directed.

The resistance threshold and test voltage are distinct quantities: Ω/V is the normalized minimum acceptance criterion; 500 V DC is the applied test voltage for the stated procedure. Do not interpret the procedure voltage as a resistance value. A Revision C sample yielded 1.8 M Ω during one validation run, comfortably above the criterion. That measured result is evidence for the tested sample and is not the specified minimum.

Revision A review notes had proposed 1,000 Ω/V as a preliminary internal target. The design review adopted the lower 500 Ω/V acceptance threshold after defining the complete test configuration and environmental conditioning. The older planning figure is not the Revision C production requirement.

6.3 Overcurrent and fault response

The module includes internal overcurrent protection and contactors. The host shall provide coordinated external protection and shall not rely on the internal fuse to protect arbitrary downstream wiring. Short-circuit testing is conducted in a controlled laboratory enclosure with remote operation and defined containment; it is not an installer test.

After a detected isolation, overcurrent, or severe thermal fault, the BMS may latch a diagnostic state. Recovery requires the host to follow the documented fault-clear sequence. Repeated key cycling or repeated charge attempts are not an acceptable fault-clearing method.

6.4 Thermal and fire precautions

Do not charge a cold module or operate beyond the thermal limits in Sections 4 and 7. Keep the module away from open flame and unapproved heat sources. A damaged, swollen, venting, or unusually hot module shall be isolated from people and adjacent equipment in accordance with the host site's emergency response plan.

The cell chemistry reduces some propagation hazards relative to other lithium-ion chemistries, but it does not make the module nonflammable or eliminate toxic decomposition products. No simplified statement of chemistry shall replace a system fire-risk assessment.

6.5 Service and end-of-life

Only trained personnel may open the module. Before enclosure removal, isolate the host, wait the specified discharge interval, and verify voltage at the service points. Replace seals and fasteners according to the controlled service procedure. A module involved in a collision, flood, or electrical fault requires inspection before reuse.

End-of-life modules shall be handled through an appropriate battery collection and recycling route. The module must not be disposed of as ordinary waste. Local requirements vary; the system owner must select a lawful route for its location.

7 Operating Conditions

7.1 Operating Environment

The general ambient operating range for the Revision C module is -20°C to 55°C . This range applies to normal operation with the module installed as intended and with any required derating observed. It is not a statement that full rated current is available at both endpoints.

Charging is inhibited below 0°C ambient. This is a charge-specific restriction within the broader operating range and reflects cell plating risk and the production BMS policy. Discharge may remain available at reduced current in cold conditions if the BMS permits it. The host shall use reported limits rather than assuming that every operation is available throughout the ambient range.

High humidity and condensation require particular care at connectors. The enclosure protection rating in Section 5.5 assumes correct assembly and does not remove the need to dry and inspect interfaces before connection. Avoid rapid transitions from cold storage into humid warm environments until condensation risk has been assessed.

7.2 Storage and Transport Conditions

For storage, maintain an ambient temperature from -30°C to 60°C with the module isolated and at the project-recommended mid-range SOC. Storage conditions are not operating conditions. Before return to service after extended storage, inspect the enclosure, confirm SOC and diagnostics, and perform the applicable incoming checks.

Transport packaging shall protect the module from impact, moisture, and terminal shorting. Shipping vibration and handling conditions are defined by the logistics plan, not by the module's operational vibration validation. A packaging study used a 5–200 Hz, 1.5 g RMS profile; that profile describes the shipping fixture study and does not replace Section 8.3's module validation profile.

During one controlled thermal exposure, a Revision B sample reached 65°C for a short interval while powered down. This was a deviation investigation, not an approved storage limit or operating condition. The sample was inspected before further use. The released storage envelope remains the one stated in this subsection.

7.3 Cooling and installation conditions

For rated continuous operation, the host shall provide coolant flow and pressure within Section 4.2 guidance and shall prevent air pockets. The module should be mounted so that the cooling base has the intended contact and the coolant ports remain accessible for inspection. The host must monitor for leakage and provide a safe response to loss of flow.

Installation clearances shall allow harness routing without bending connector backshells or pinching insulation. Keep service access available for isolation and inspection. Enclosure mounting must not obstruct pressure relief features or apply concentrated loads to the top cover.

7.4 Altitude, vibration, and contamination

The design basis assumes non-condensing indoor or protected-vehicle use. Sustained exposure to conductive dust, salt spray, aggressive cleaning chemicals, or unfiltered abrasive particles is outside the baseline qualification unless separately assessed. Altitude-related cooling and insulation effects must be reviewed for installations substantially above the project design elevation.

Road shock and host-structure modes can amplify local loads. Integrators should compare their vibration environment with the validated profile in Section 8.3 and conduct system-level testing where mounting stiffness or accessory mass differs materially.

8 Testing & Validation

8.1 Validation approach

Validation used a combination of inspection, electrical characterization, thermal mapping, mechanical testing, environmental exposure, and endurance cycling. Procedures define sample configuration, instrumentation, stabilization, acceptance criteria, and deviations. Test results in this chapter are evidence for the specific units and procedures listed; they are not universal guarantees about every manufactured module.

Revision C samples were drawn from production-intent builds. Where a procedure used a Revision B prototype, the result is explicitly identified and its applicability is limited to the unchanged design feature. The test archive records raw data, calibration certificates, photographs, and signed run sheets under controlled fictional project record numbers.

8.2 Electrical characterization

Electrical characterization verified polarity, contactor operation, communication status, and current measurement against calibrated instruments. At room temperature, the sample terminal voltage at the nominal SOC reference was 50.9 V. That is a measured operating point and should not be substituted for the design nominal voltage in Section 3.1.

The production current-limit check confirmed that the BMS reported the expected allowable-current state under the test setup. The transient sequence was applied only after thermal stabilization and included a cooldown interval. No test procedure authorizes the host to disregard real-time BMS limits.

8.3 Vibration Validation

The Revision C mechanical validation profile was a 10–500 Hz random vibration input at 3.0 g RMS for 8 hours per axis. The module was mounted through its production attachment points with representative harness restraint. Pre-test and post-test inspections checked fastener witness marks, connector retention, enclosure condition, and electrical continuity.

Axis	Exposure	Observation
X	8 h at specified profile	No structural damage observed; fastener torque marks remained aligned
Y	8 h at specified profile	Minor harness witness marks; routing restraint adjusted for subsequent builds
Z	8 h at specified profile	No loss of electrical continuity; post-test inspection passed

The Revision B prototype campaign ran for 5 hours per axis using a different fixture resonance correction. Those data supported early design learning but do not define the Revision C validation duration. The host's shipping profile in Section 7.2 likewise addresses packaging exposure rather than mounted module vibration qualification.

8.4 Capacity and thermal characterization

Capacity characterization used a controlled chamber, a defined discharge current, and endpoint rules tied to the module's normal SOC window. One Revision C sample delivered 5.06 kWh during the recorded run. Cell temperature remained below the applicable operational limit, and the result was within the laboratory's expected spread for the test condition. Rated nominal energy is specified in Section 3.2; this measurement is not a new rating.

Thermal mapping at a steady load recorded a maximum 6.2 °C cell-to-cell spread after stabilization. The test used the production cooling base and the coolant flow called out in Section 4.2. The acceptance requirement

is stated in Section 4.4. Separate Revision B mapping observed a wider spread and led to a flow-path modification before the C release.

8.5 Cycle-Life Acceptance

The production cycle-life acceptance criterion is at least 2,000 equivalent full cycles before measured capacity falls below 80% of initial capacity, using the defined reference profile and controlled temperature. EFC accounting normalizes partial cycles by total discharge throughput; it does not mean that each test cycle must traverse the full SOC window.

At the date of this report, the longest-running Revision C endurance sample had completed 1,250 EFC and retained 91% of its initial measured capacity. The run was ongoing and had not reached the acceptance endpoint. An early planning presentation projected a 3,000 EFC design objective; that aspirational target was not adopted as a release acceptance criterion. The current formal criterion remains in this subsection.

The acceptance test controls charge rate, discharge rate, temperature, rest intervals, and capacity measurement method. Changes to any of these parameters require a comparability assessment. A single sample's interim retention cannot establish population reliability or service life.

8.6 Environmental and ingress checks

Ingress testing was performed on the assembled Revision C enclosure with production seals and closures. The tested unit passed the defined IP67 exposure sequence and showed no visible internal water after inspection. The result applies to the test configuration and does not certify a field-repaired enclosure with unverified seals.

Thermal cycling and humidity exposure were conducted on separate samples. Connector contacts were inspected for corrosion and retention after conditioning. A coolant compatibility soak evaluated seal material coupons; it did not qualify every possible host coolant formulation.

8.7 Deviations and evidence limitations

One vibration run paused to correct a fixture accelerometer orientation before the full exposure was accumulated. The corrected run was repeated and only the valid exposure was used in the summary. Raw records preserve both runs. The validation summary does not claim statistically meaningful production reliability from the limited sample count.

Revision B test results are identified in the record set and are not automatically transferable to Revision C. The project review considered each unchanged feature and documented the rationale. Future design changes to busbars, enclosure seals, controller firmware, or cooling geometry may require partial or full revalidation.

9 Limitations & Known Issues

9.1 Intended-use limitations

The BM-51 has not been represented as qualified for aircraft, medical equipment, life-safety systems, explosive atmospheres, or unattended use without host monitoring. The fictional validation program does not substitute for product certification. The integrator shall assess the complete system and intended operating jurisdiction.

Parallel operation of multiple modules requires current-sharing analysis, compatible firmware, and a system-level fault strategy. This report does not authorize arbitrary series or parallel combinations. Use beyond the Section 7 conditions requires engineering review and additional evidence.

9.2 SOC Estimation Transient

On the Revision C production configuration, the displayed capacity indication may temporarily lag actual capacity by up to 4 percentage points after rapid load changes. The estimator converges as current and voltage conditions stabilize. The behavior can affect the smoothness of the displayed SOC but does not affect protection limits, cell monitoring, or independent BMS fault response.

Host applications should avoid treating a brief SOC display change as a direct measurement of energy loss. For control decisions, use the module's allowable-current and fault signals. This issue is tracked as a software estimation limitation and does not alter the specified SOC window in Section 3.2.

9.3 Superseded prototype indication behavior

Revision B prototypes exhibited a separate estimator behavior: under repeated high-current steps, the displayed SOC could lag by as much as 7 percentage points and remain unsettled until a longer rest interval. That behavior was associated with the earlier calibration and sensing harness. It is not the Revision C issue described in Section 9.2 and must not be applied to the current production configuration.

9.4 Other known limitations

The BMS SOC estimate is less stable at very low current when the module has experienced a wide temperature gradient. Capacity should be assessed using the controlled procedure rather than a dashboard snapshot. Sensor placement does not capture every local hot spot, so thermal margin and proper cooling remain necessary.

The module reports diagnostics over the host communication interface but does not provide a complete prognostic health assessment. Maintenance intervals and replacement decisions require system-level usage data and an approved service process. Field data collection shall respect applicable privacy and cybersecurity controls in the host product.

9.5 Open actions

Action	Status at Revision C release	Closure evidence
Refine SOC settling characterization over aging	Open, monitoring	Additional endurance data and estimator review
Confirm coolant seal compatibility for alternate glycol blends	Open, host-specific	Material compatibility test for selected coolant
Expand vibration sample count	Planned	Additional production-intent builds

Open actions are not waivers of mandatory requirements. Any integration that depends on an open action must document the applicable risk and verification plan.

10 Revision History

10.1 Revision hierarchy

Revision sequence is chronological and configuration-specific. Later revisions supersede earlier values only where the change record identifies the affected specification. The Revision C clauses in this report define the current production release. Historical data remain visible to explain design evolution and are not normative for Revision C unless explicitly adopted.

10.2 Change record

Revision	Date	Configuration	Summary of change	Disposition
A	2025-02-12	Early design	Preliminary energy concept, initial enclosure envelope, and draft isolation objective	Not released; planning material only
B	2025-09-30	Prototype / validation	Updated cell arrangement, experimental current settings, prototype cooling map, and first estimator calibration	Engineering prototype; not production-released
C	2026-09-15	Current production release	Production busbar and sensing harness, revised thermal base, controlled enclosure seals, and final released limits and acceptance criteria	Current authoritative configuration

10.3 Change applicability notes

Revision C adopted the production thermal flow path after Revision B mapping identified excessive cell temperature spread. The controller calibration and sensing harness were also updated, so estimator behavior from Revision B is not directly applicable. Mechanical exposure data were reviewed by feature; test applicability is recorded in the validation archive rather than inferred solely from a shared part name.

Revision C release approval applies to the configuration identifier shown on the rating label and its associated controlled bill of materials. A deviation, alternate component, or firmware replacement creates a configuration difference that must be reviewed before the limits in this report are assumed to remain valid.

Appendix A. Interpretation of values and evidence

A.1 Specification versus result

A specification defines the design commitment or acceptance boundary under named conditions. A test result records what a particular specimen did in a particular procedure. For example, a sample may exceed a minimum resistance requirement by a large margin, but the measured result does not become the new minimum requirement. Similarly, an interim endurance result describes progress, not completion of the cycle-life acceptance test.

A.2 Conditions attached to limits

Temperature, SOC, current duration, coolant flow, mounting, sealing state, and revision can change how a number should be interpreted. Where a value appears without its full condition nearby, use the subsection

reference and the applicable conditions in this report. Do not combine a storage envelope with an operating rating or apply a component rating to the assembled module.

A.3 Record traceability

Each validation record includes the specimen revision, serial identifier, procedure revision, instrumentation, calibration status, environmental conditions, and deviations. Test summaries in this report are abbreviated for engineering review. The controlled project archive is the source for detailed waveforms and raw measurements; the archive is also fictional and is not required to interpret the requirements stated here.